UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 3 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Find the equation of the straight line passing through (0, -5) and (-2, 4).

(4 marks)

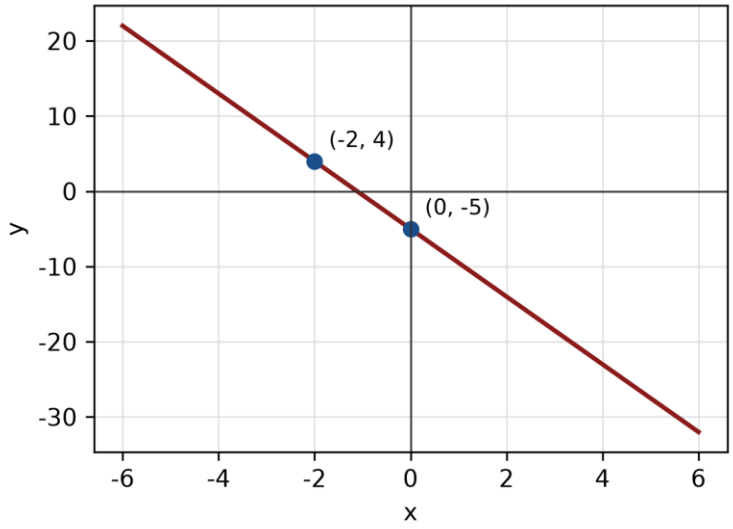


Figure: line through the two points.

b) Find the equation of the line passing through (-2, 1) and parallel to the line -4x + 5y + 3 = 0.

(5 marks)

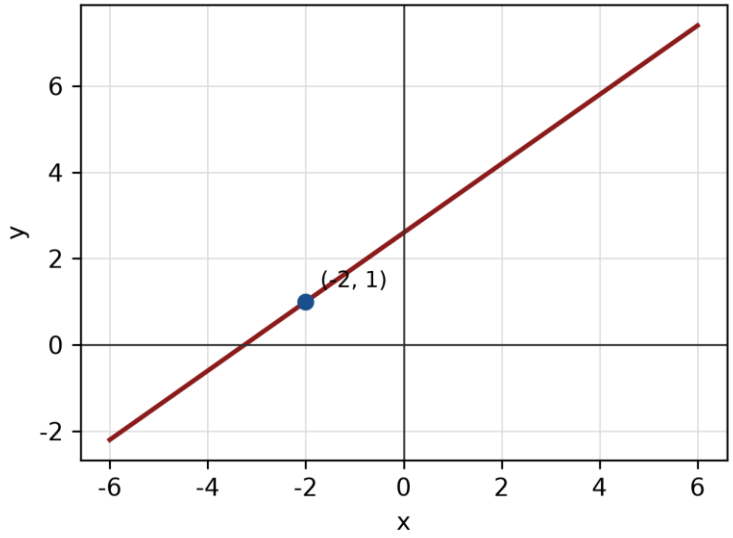


Figure: required line.

c) Find the equation of the line with x-intercept 8 and y-intercept -8.

(4 marks)

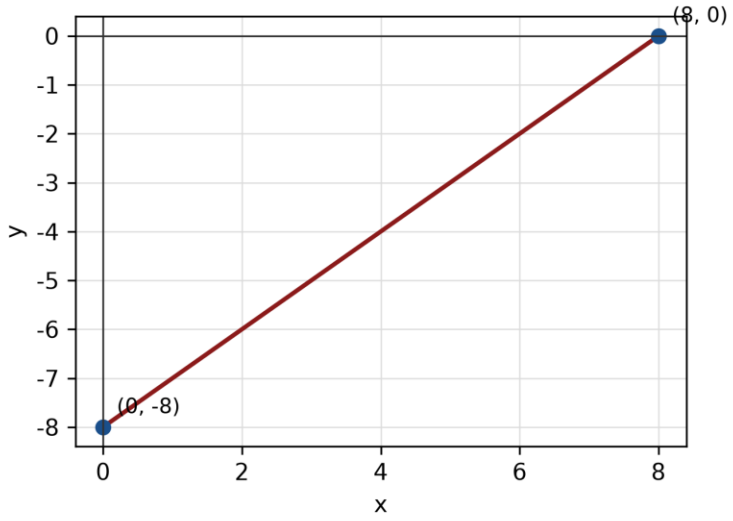


Figure: line and intercepts.

QUESTION 2

a) Write the equation of the circle with centre (-5, 4) and radius 3. Hence write its general equation and sketch the circle.

(5 marks)

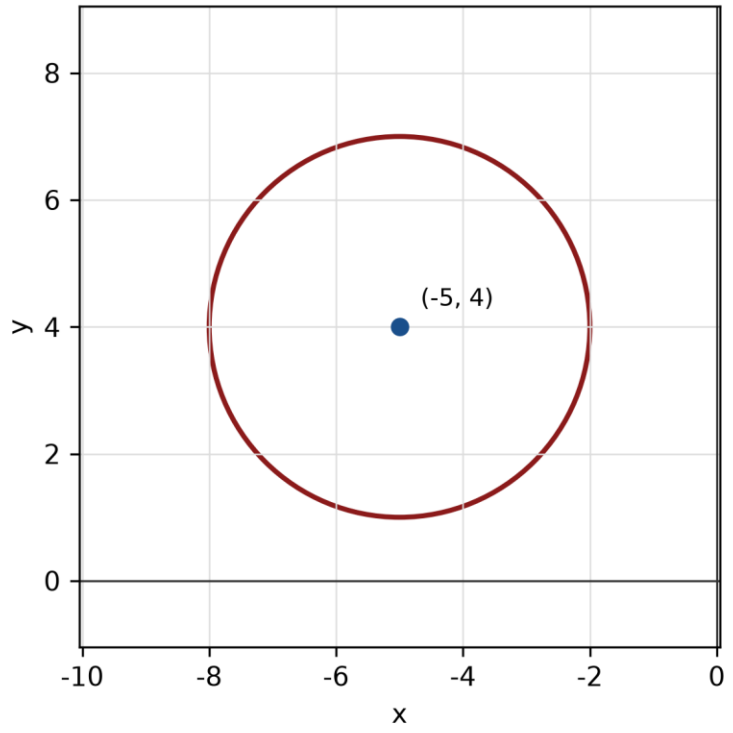


Figure: circle.

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b) Write $x^2 + 8x + y^2 - 10y + 25 = 0$ in standard form. Hence state the centre and radius and sketch the circle.

(5 marks)

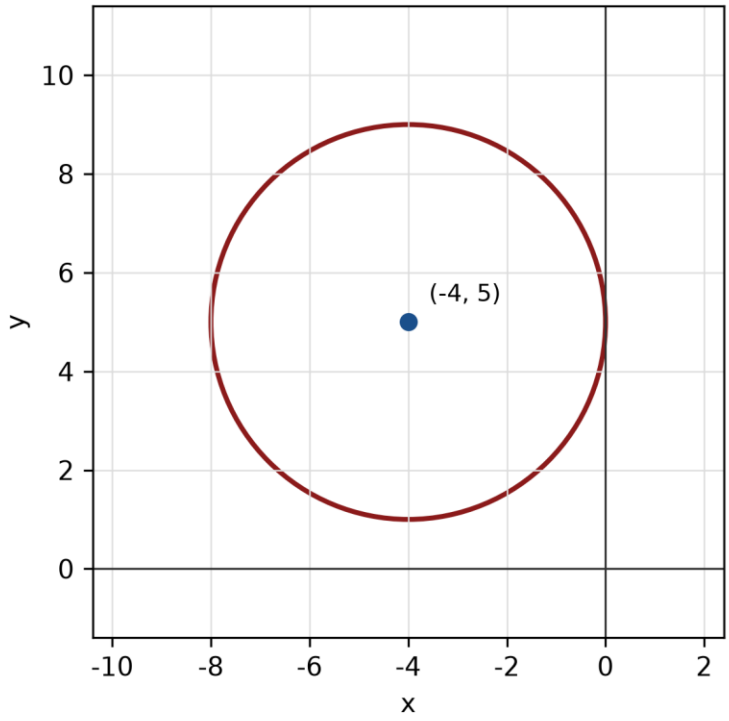


Figure: circle.

c) The endpoints of a diameter of a circle are (2, 2) and (-6, -5). Find the centre, radius and equation of the circle.

(5 marks)

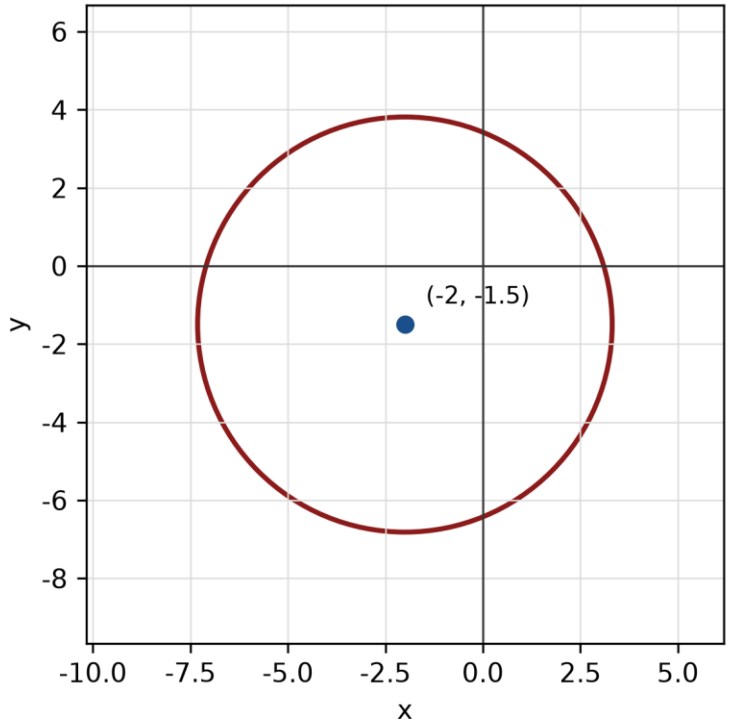


Figure: circle from diameter.

d) Sketch the circle $x^2 - 10x + y^2 + 8y - 8 = 0$. State its centre and radius.

(4 marks)

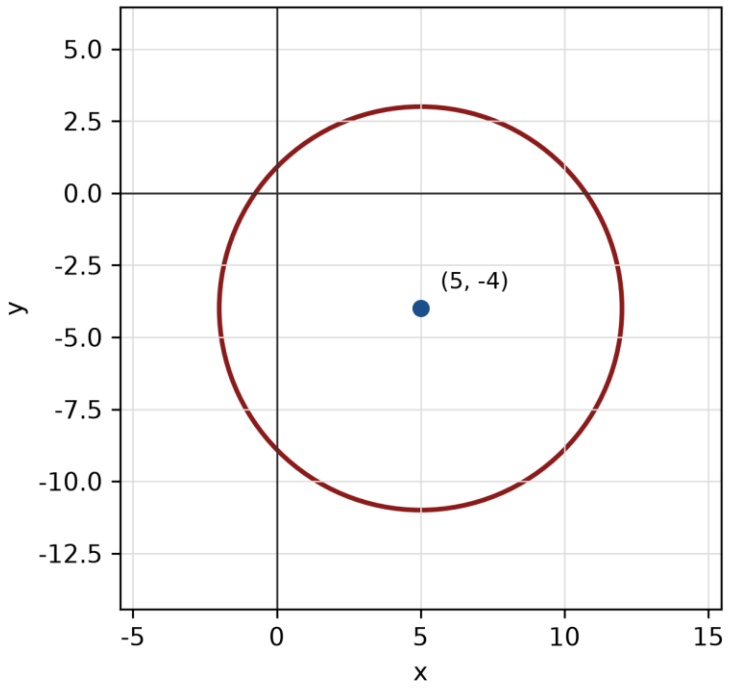


Figure: circle.

QUESTION 3

a) Write $y = x^2/4 - x/2 - 19/4$ in standard form. Hence state the vertex, focus and directrix, and sketch the parabola.

(5 marks)

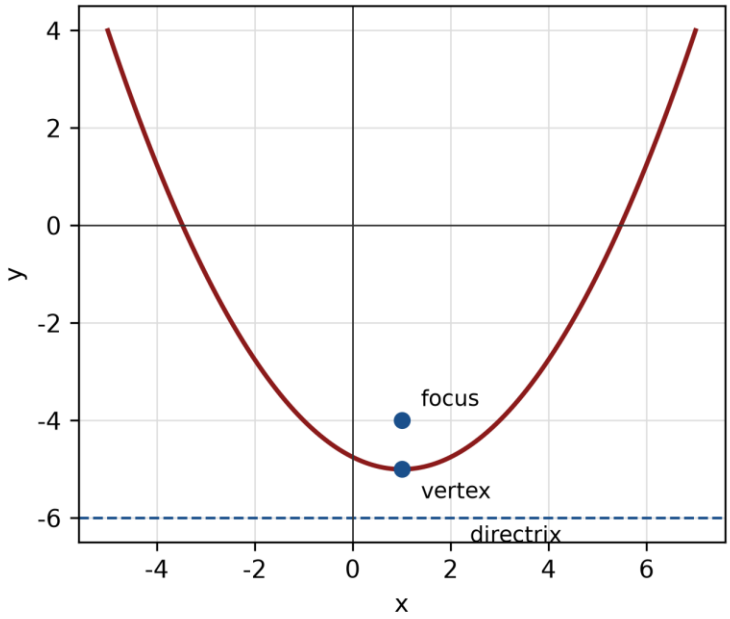


Figure: parabola.

b) Find the standard equation of the parabola with focus (1, 6) and directrix y = 4. State its vertex.

(5 marks)

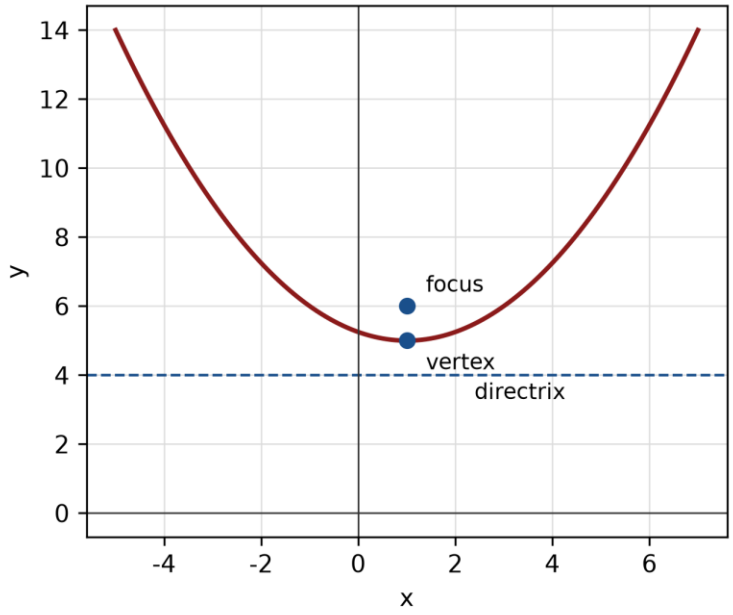


Figure: parabola with focus and directrix.

c) Sketch the parabola $(x + 3)^2 = 4(y - 5)$. Label the vertex, focus and directrix.

(5 marks)

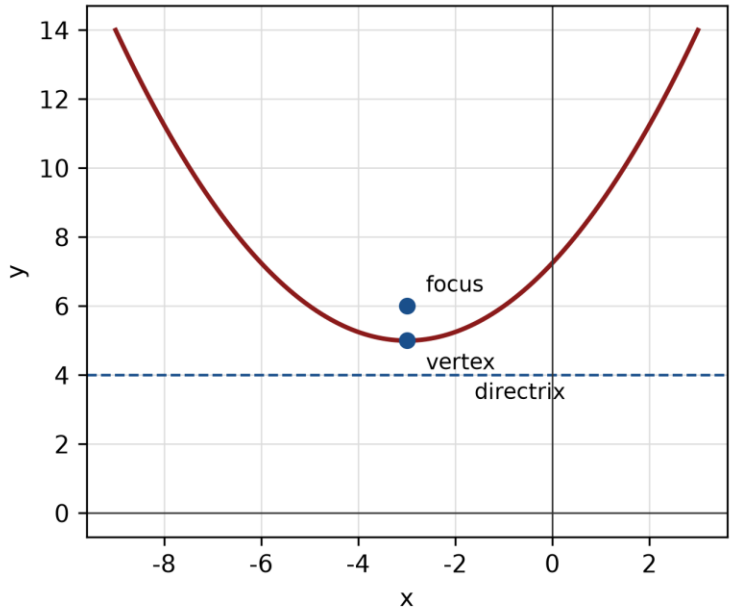


Figure: parabola.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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